

Student Rubric

DS 4002 - Instructor: Javier Rasero

Case Study Topic: Trump Tariff Announcements and the S&P500

Why am I doing this?

You are completing this case study to practice principals of data science and statistics using real world data. You will learn how to transfer data from a tsv to a Pandas dataframe, filter the data, extract financial data, and run statistical tests in Python. You will learn to support a claim with clear visuals and results.

What am I going to do?

You will use the provided notebook and data instructions to:

1. Load and filter Trump social media posts to focus on tariff posts.
2. Extract financial data for the timespan that you have data for Trump's posts and calculate important numbers.
3. Conduct statistical tests to form conclusions.
4. Create visuals to help a reader understand your conclusions.
5. Write a succinct article draft sharing your findings.

How will I know I have succeeded?

You will know you have succeeded if: your python notebook runs in one go without any errors; you are able to calculate important financial statistics; you have reached a conclusion on the impact of Trump's tariff announcements; you are able to put this conclusion in context and report on its limitations.

Article Formatting	• Structure and Flow: The article follows a logical progression, including a clear introduction, methodology, analysis, and conclusion. • Markdown Proficiency: Proper use of headers, bold text, and lists to enhance readability and organize complex information. • Clarity and Tone: Technical concepts are explained in an accessible manner, maintaining a professional yet engaging narrative. • Visualization: At least two figures are included in the article

Python Code	• Code Efficiency: Implementation of optimized logic and built-in functions to handle data processing without redundant computations. • Readability: Adherence to standard naming conventions, consistent indentation, and the use of meaningful variable names. • Documentation: Inclusion of inline comments and docstrings that explain the "why" behind specific code blocks.
Statistical Analysis	• Methodological Rigor: Selection of statistical tests that are appropriate for the data and research question. • Metric Accuracy: Correct calculation of key indicators such as p-values, confidence intervals, and effect sizes. • Conclusions: Identify whether Trump's tariff announcements on social media influence intraday market volatility and daily returns. All conclusions made are supported by statistical analysis. • Data Cleaning: Transparent documentation of how outliers and missing values were handled to prevent bias.
Results Visualization	• Visual Clarity: Charts and graphs are clean, with legible axes, clear labels, and intuitive legends. • Plot Selection: Choice of visualization types (e.g., heatmaps for correlation, boxplots for distribution) that best represent the underlying data patterns. • Insight Integration: Visuals are directly referenced within the text to support specific claims rather than standing as isolated images. • Aesthetic Consistency: Use of a professional color palette and consistent scaling across all figures.
Analysis Contextualization	• Interpretation of Findings: Moving beyond raw numbers to explain what the statistical results actually mean in a real-world or theoretical context. • Limitation Acknowledgment: Honest assessment of potential biases, sample size constraints, or variables not captured in the study (intraday volatility). • Actionable Insights: Synthesis of data into "so-what" conclusions based on the observed trends. • Connection to Objectives: Explicitly linking the results back to the primary goal of the article.